

Predictive Model Plan – Student Template

Use this template to structure your submission. You can copy and paste content from GenAI tools and build around it with your own analysis.

1. Model Logic (Generated with GenAI)

Use a GenAI tool (e.g., ChatGPT, Gemini) to generate the logic or structure of your predictive model.

- You may include pseudo-code, a step-by-step process, or a simplified code snippet.
- Briefly explain what the model is designed to do.

Paste your GenAI-generated output below or describe the logic in your own words:

Recommended approach: a two-stage modeling strategy.

Stage 0 – Target validation (prerequisite): Before building any model, re-confirm with Geldium how Delinquent_Account was defined and labeled. Our EDA found that none of the available input variables (Credit_Utilization, Missed_Payments, Credit_Score, Debt-to-Income Ratio, Income, Account_Tenure, or 6-month payment history) correlate meaningfully with this target (all coefficients between -0.04 and +0.05). This is far below industry expectation and should be resolved before model results are trusted operationally.

Stage 1 – Baseline model: Logistic Regression

Pipeline:

1. Data ingestion: Load cleaned, imputed dataset (median imputation for Income/Loan_Balance, regression imputation for Credit_Score, standardized Employment_Status categories).
2. Feature engineering: Encode categorical variables (Employment_Status, Credit_Card_Type, Location) using one-hot encoding; engineer a 6-month payment behavior score (e.g., weighted count of Missed/Late statuses) as a candidate feature.
3. Train/test split: 70/30 split, stratified by Delinquent_Account to preserve the 84%/16% class balance.
4. Model training: Fit a logistic regression model with L2 regularization; output is a probability score (0–1) representing delinquency risk.
5. Thresholding: Convert probability to a risk tier (Low / Medium / High) using business-defined cutoffs rather than a single 0.5 threshold, to support tiered intervention strategies.
6. Output: Risk score and tier per customer, feeding into the Collections team's prioritization queue.

Stage 2 – Comparison model: Decision Tree / Gradient Boosted Trees (e.g., XGBoost)

Used as a benchmark against the logistic regression baseline to check whether non-linear feature interactions (e.g., combinations of utilization + tenure + employment status) capture signal that a linear model misses. If accuracy gains are modest, the simpler logistic regression should be preferred for deployment given its interpretability advantage (see Section 2).

Top 5 candidate input features (pending Stage 0 validation):

1. Credit_Utilization – proportion of available credit in use
2. Missed_Payments – historical count of missed payments
3. Debt_to_Income_Ratio – financial strain indicator
4. Credit_Score – standard underwriting risk signal
5. Engineered 6-month payment behavior score (from Month_1–Month_6 history)

Note: these were selected based on standard industry practice and the variables specified in Geldium's dataset; however, our EDA found weak observed correlation with the current target variable. We recommend treating this feature list as a starting hypothesis to be re-tested once Stage 0 is resolved, rather than as a confirmed predictive set.

2. Justification for Model Choice

Explain why you selected this specific model type (e.g., logistic regression, decision tree, neural network). Consider:

- Accuracy
- Transparency
- Ease of use or implementation
- Relevance for financial prediction
- Suitability for Geldium's business needs

Logistic regression is the recommended primary model for Geldium's delinquency prediction system. In regulated financial services, model transparency is not optional — Geldium's Collections team and compliance function need to explain, to both internal stakeholders and potentially regulators or customers, why a specific individual was flagged as high-risk. Logistic regression produces coefficients that map directly to each feature's contribution to risk, making it straightforward to generate plain-language explanations (e.g., “a 10-point drop in credit score increases predicted risk by X%”) — something far harder to do with a neural network and only achievable in tree-based models through additional tooling like SHAP. While gradient-boosted trees or neural networks may offer marginal accuracy gains by capturing non-linear feature interactions, that gain must be weighed against the loss of interpretability, the added complexity of deployment and ongoing monitoring, and the heightened regulatory scrutiny financial

institutions face when using “black box” models for credit decisions. Given Geldium's stated goals — supporting human-reviewed Collections decisions, ensuring fairness, and maintaining transparency — logistic regression strikes the right balance between predictive utility and accountability, with a tree-based model retained only as a benchmarking comparison rather than the production system.

3. Evaluation Strategy

Outline how you would evaluate your model's performance. Include:

- Which metrics you would use (e.g., accuracy, precision, recall, F1 score, AUC)
- How you would interpret those metrics
- Any plans to detect or reduce bias in your model
- Ethical considerations in making predictions about customer financial behavior

Given the 84%/16% class imbalance (only 16% of customers are delinquent), accuracy alone is a misleading metric — a model that always predicts “not delinquent” would score 84% accuracy while being useless. The evaluation strategy therefore prioritizes:

Accuracy & reliability metrics:

- *Precision – of customers flagged as high-risk, what proportion actually become delinquent. Important for avoiding wasted Collections outreach on false positives.*
- *Recall (Sensitivity) – of all customers who become delinquent, what proportion the model successfully flags. Critical given the business cost of missing a genuinely at-risk customer.*
- *F1 Score – harmonic mean of precision and recall, used as the primary single-number metric given the class imbalance.*
- *AUC-ROC – measures the model's ability to rank customers by risk across all thresholds, useful for tuning the risk-tier cutoffs (Low/Medium/High).*
- *Confusion Matrix – reviewed each evaluation cycle to understand the specific trade-off between false positives (unnecessary outreach) and false negatives (missed risk).*

Fairness & bias checks:

- *Disparate impact analysis – compare delinquency flag rates across Age groups, Employment_Status, and Location to check no group is disproportionately flagged as high-risk relative to its actual delinquency rate.*
- *Demographic parity / equal opportunity metrics – evaluate whether true positive rates (recall) are consistent across customer segments, not just overall accuracy.*
- *Proxy variable review – Location and Credit_Card_Type should be reviewed to ensure they are not acting as proxies for protected characteristics (e.g., socioeconomic or demographic patterns tied to geography).*

Interpreting results & deciding on improvement:

- *If F1 score and AUC are low across the board (as our EDA correlation findings suggest is likely until the target variable is validated), this signals a data/labeling issue rather than a model architecture issue — more complex models will not fix a weak signal-to-target relationship.*
- *If fairness metrics show meaningful disparity across groups, the model should not be deployed until bias mitigation techniques (e.g., reweighting training samples, fairness-constrained optimization, or removing/adjusting proxy variables) are applied and re-tested.*
- *Models should be re-evaluated on a regular cadence (e.g., quarterly) as new repayment data becomes available, since customer risk behavior and macroeconomic conditions shift over time.*

Ethical considerations:

- *All risk scores should support, not replace, human judgment in Collections decisions — the model output is a prioritization signal, not an automatic action trigger.*
- *Customers should have a path to contest or seek clarification on a risk-based intervention, consistent with responsible AI principles around transparency and recourse.*
- *Given our Task 1 finding that current features show weak correlation with the target, deploying a model before resolving this issue risks delivering misleading risk scores to the Collections team — an outcome that could harm both business outcomes and customer trust.*